Task 9 – IAM User Creation with Restricted EC2 Access (Region-Locked to Mumbai)

Objective

Create an IAM user from the Root AWS account.
Set the policy or permission to the user, so that it can only access AWS EC2 service.
Restricted privileges.
Login as the IAM user and verify the working of the policies attached with the user.

Note: try the following

IAM user with AWS Management Console access
IAM user with Programmatic access

Requirements

- AWS root account access
- IAM service
- AWS Management Console
- AWS CLI installed on a local machine
- A custom IAM policy with region-based conditions

Procedure

Step 1 – Create the IAM User (from Root Account)

While logged in as the AWS root user, a new IAM user was created:

- IAM → Users → Create user
- Username: john
- Access type selected: "Provide user access to the AWS Management Console"
- A custom password was set for the user

Step 2 – Create a Custom Restricted Policy

Instead of using the AWS-managed AmazonEC2FullAccess policy (which grants EC2 access across all regions), a custom policy was created to restrict EC2 access specifically to the Mumbai region.

IAM → Policies → Create policy → JSON tab

json
```
{
   "Version": "2012-10-17",
   "Statement": [
      {
         "Sid": "AllowEC2ActionsInMumbai",
```

```
      "Effect": "Allow",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:RequestedRegion": "ap-south-1"
        }
      }
    },
    {
      "Sid": "RequiredGlobalEC2ViewActions",
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeRegions"
      ],
      "Resource": "*"
    }
  ]
}
```

- The first statement allows all EC2 actions, but only when the request's region matches ap-south-1 (Mumbai).
- The second statement allows the ec2:DescribeRegions action globally, since this is required for the AWS Console's region selector to function correctly, and does not grant access to any actual EC2 resources outside Mumbai.

The policy was named EC2-Mumbai-Only and created successfully.



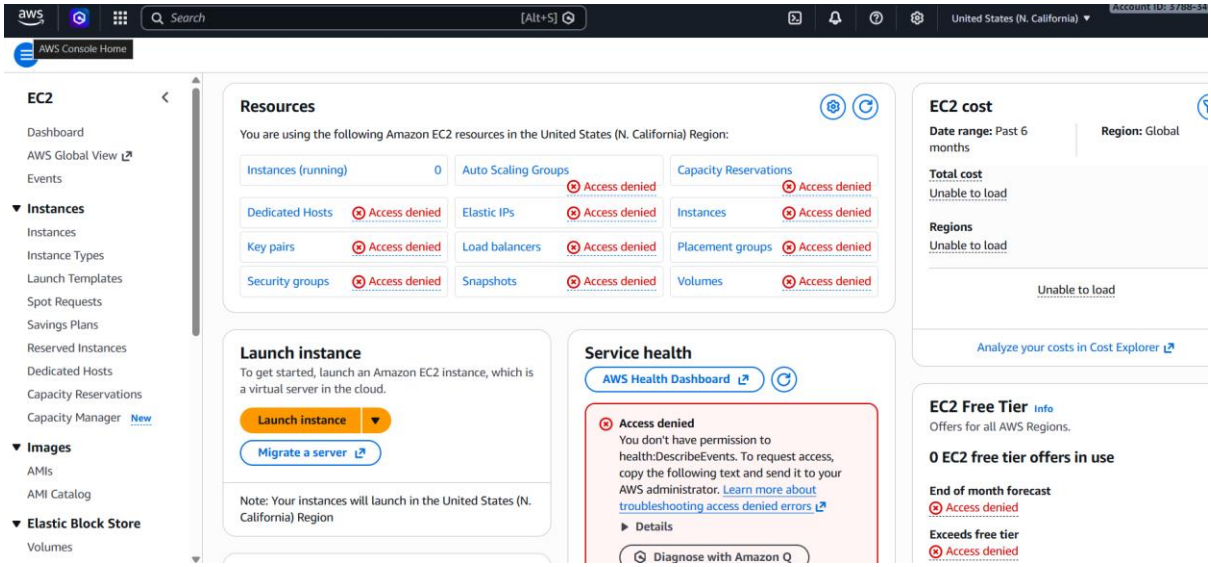
Step 3 – Attach the Policy to the IAM User

The custom policy (EC2-Mumbai-Only) was attached directly to the user john under the Permissions section

Step 4 – Verify Using AWS Management Console Access

The user logged out of the root account and logged in as john using the account-specific IAM sign-in URL.

- While the console region was set to a region other than Mumbai (e.g., United States - N. California), attempting to view EC2 resources resulted in "Access denied" messages across Instances, Security groups, Volumes, Key pairs, and other EC2-related resources.
- Switching the console region to Asia Pacific (Mumbai) ap-south-1 allowed EC2 resources to be viewed successfully, confirming the restriction was correctly enforced through the browser-based console.



Step 5 – Generate Access Keys for Programmatic Access

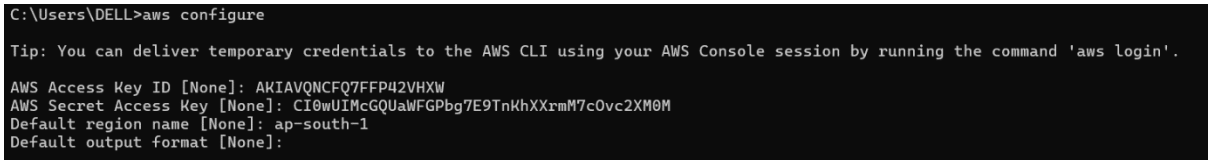
To test Programmatic access, an access key was generated for the same user:

- IAM → Users → john → Security credentials tab → Create access key
- Use case selected: Command Line Interface (CLI)
- The Access Key ID and Secret Access Key were saved securely.

Step 6 – Configure AWS CLI

The AWS CLI was installed and configured on a local Windows machine using john's credentials:

aws configure
AWS Access Key ID: <john's access key>
AWS Secret Access Key: <john's secret key>
Default region name: us-east-1
Default output format: json



Step 7 – Verify Using Programmatic Access (AWS CLI)

Testing access outside the permitted region:

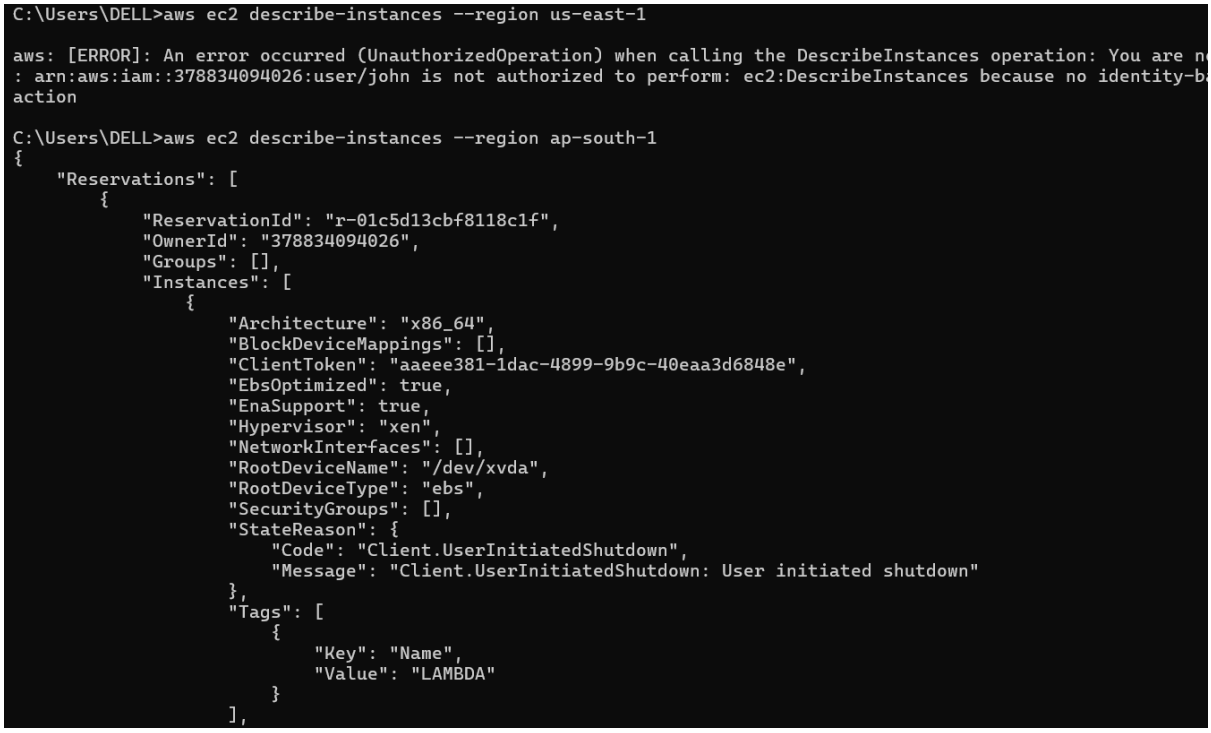
aws ec2 describe-instances --region us-east-1

Result:

An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform: ec2:DescribeInstances because no identity-based policy allows this action.

Testing access within the permitted region:

aws ec2 describe-instances --region ap-south-1



Result: A successful response was returned, listing EC2 instance details (Reservation ID, Owner ID, instance architecture, tags, and state information), confirming that access was correctly granted in the Mumbai region.

Verification

- An IAM user (john) was successfully created from the root AWS account.
- A custom policy was created and attached, restricting the user to EC2 actions only within the Mumbai (ap-south-1) region.
- Console access was tested: EC2 access was denied in a non-Mumbai region and successfully allowed after switching to the Mumbai region.
- Access keys were generated for the same user to enable Programmatic access.
- Programmatic access was tested via AWS CLI: EC2 access was denied when targeting us-east-1 and successfully allowed when targeting ap-south-1.
- Both access methods (Console and Programmatic) confirmed the same restricted behavior, validating that the IAM policy was correctly enforced regardless of access method.

Conclusion

An IAM user was successfully created from the AWS root account with tightly restricted permissions, limited specifically to EC2 actions within the Mumbai region. A custom IAM policy using the aws:RequestedRegion condition was used to enforce this region-based restriction, with an additional global exception for the DescribeRegions action to preserve console usability. The restriction was verified through two independent access methods — AWS Management Console access and Programmatic access via the AWS CLI — with both methods consistently denying EC2 actions outside Mumbai and allowing them within Mumbai. This task provided practical experience with IAM user management, the principle

of least privilege, custom policy authoring using conditions, and verifying access controls across multiple authentication methods.